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Dimos Chatzinikolaou and Charis Vlados

Dimos Chatzinikolaou, University of Nicosia, School of Business. https://chdimosthenis.github.io/

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Crisis, innovation and change management: A blind spot for micro-firms?

Dimos Chatzinikolaou (corresponding)[a], dimchatz@econ.duth.gr

a. Department of Economics, Democritus University of Thrace, Komotini, Greece, 69100

Charis Vlados[a,b,c], cvlados@econ.duth.gr

a. Department of Economics, Democritus University of Thrace, Komotini, Greece, 69100
b. School of Business, University of Nicosia, Nicosia, 46 Makedonitissas Avenue, CY-2417, PO Box 24005, Cyprus
c. School of Social Sciences, Business & Organisation Administration, Hellenic Open University, Par. Aristotelous 18, Patras, Greece, 26335

Abstract

Purpose

This paper aims to explore how the owners of less competitive micro-firms perceive the "crisis–innovation–change management" triangle. It examines whether their understanding of these overarching entrepreneurship theory principles is inadequate compared to the relevant scientific literature.

Design/methodology/approach

This qualitative analysis follows principles based on the inductive method and grounded theory, thickly describing the results from research conducted in a sample of 38 tertiary-sector micro-firms in the Eastern Macedonia and Thrace region—one of the least developed and competitive areas across Europe. It triangulates the data with 11 respective small firms.

Findings

Micro-firm owners perceive the crisis as an ostensibly exogenous phenomenon, innovation as something quasi-unattainable—although vaguely significant—and change management as a relatively unknown process. This understanding lies somewhat distant from the extant literature that examines the structural nature of crises, the innovational power to exit profound restructurings, and the rebalancing requisite for building new overall organizational methods to

survive this internal-external transformation. In essence, the triangle crisis–innovation–change management is a blind spot for the examined micro-firm owners as they ignore its significance as an adaptation mechanism—contrary to several direct competitors.

Social implications

Based on the reluctance of these individuals to cultivate their systematic business knowledge, it seems unrealistic that they would seek to pay the necessary high price for business consulting in the future. An ideal solution would be to build public entrepreneurship clinics to provide these less dynamic and adaptable organizations with free preliminary or in-depth counseling. The Institute of Local Development–Innovation (ILDI) could aim to provide free consulting services to reinforce organizational physiology by coordinating different socioeconomic actors.

Originality

To the best of our knowledge, this empirical research is one of the first to test the comprehension of weaker micro-firms—less competitive and developed in organizational terms—to the triangle crisis–innovation–change management.

Keywords: entrepreneurship, micro-firms, crisis, innovation, change management, entrepreneurship reinforcement policy

Plain language summary

Micro-firm owners perceive the crisis as an ostensibly exogenous phenomenon, innovation as something quasi-unattainable—although vaguely significant—and change management as a relatively unknown process. These micro-firm owners in less developed ecosystems are unaware of the "crisis–innovation–change management" triangle as an adaptation mechanism. The Institute of Local Development–Innovation (ILDI) could aim to provide free consulting services to reinforce these firms' organizational physiology by coordinating different socioeconomic actors.

1. Introduction

The strategic management literature focuses mainly on the dynamic small and medium-sized enterprises (SMEs) and startups when referring to organizations of smaller size and entrepreneurial breadth (Blank, 2013)—it usually pays less attention to micro-firm competitiveness issues. According to the widely used definition by the European Commission (2020), micro-firms employ fewer than 10 people and generate revenues of up to 2 million. Also, small firms have fewer than 50 people, with a turnover of up to 10 million. Recent studies show that micro-firms are structurally differentiated due to quality characteristics and specificities (Farè, 2021; Granata *et al*., 2018). Some approaches emphasize components primarily traced in larger firms, such as R&D and green innovation (Baumann and Kritikos, 2016; Mellett *et al*., 2018; Yıldırım & Arun, 2019). Also, other related research shows that micro-firms can be equally innovative as their small and medium counterparts (Slavec, 2022) and that innovation is not a systematic process in micro-firms due to their lack of specific resources (Faherty and Stephens, 2016).

Over the past decade, some interesting conclusions have emerged by investigating specific micro-firm innovation issues. By searching in related high-impact journals for studies in micro-firms, it can be discerned that the primary interest and findings converge in identifying particular factors that favor or hinder the emergence of innovation. Table 1 presents specific contributions to what innovativeness problems arise in such organizations. The lack of resources and the unfavorable location seem to be the two main factors that make a micro-firm relatively weak.

Micro-firm innovation	**Open issues**
Rigg *et al*. (2021, pp. 621, 636)	
"*For micro-firms, with even more profound restrictions on time, resources and capabilities, systematic innovation is rare.*"	Peripheral locations can hinder micro-firms from accessing knowledge resources.
Berrone *et al*. (2014, pp. 477, 496)	
"*… human capital (proxied by educational level and degree of dedication), innovation, and intensity of use of own capital are important determinants of the microenterprise's performance.*"	Policymakers must channel funds to people capable of handling an enterprise, a trait rarely present in micro-firms.
Faherty and Stephens (2016, pp. 349, 350)	
"*… all of the micro enterprises studied engage in a range of innovation activities across products,*	Barriers to innovation include limited time and capacity, short-sighted operational focus, absence of

processes, people and marketing. … however, innovation is not a managed or systematic process, and this is often due to lack of resources."	consulting structures, a lack of qualified personnel, shortage of finance, inaccessibility or fear of partners, dominant competitors, and lack of opportunity awareness.
Lin *et al*. (2020, pp. 1, 4)	
"*Although technological innovation gives micro-enterprises a competitive advantage to survive in business competition, in the general economy these businesses hardly can compete with larger organizations based solely on their knowledge and value.*"	Micro-firms need collaborative innovation by merging forces with others, improving existing products with interactive channels, and involving key suppliers or customers.

Table 1. Recent contributions to micro-firm innovation.

Micro-firms appear to face problems in generating progress through continuous and gradual innovation. Entrepreneurial capabilities, the firm's location, and the related entrepreneurship reinforcement policies are among the main open issues for innovation generation in micro-firms. According to Buldyrev *et al*. (2020), personalized products and targeting small markets (e.g., niche marketing) can lead to innovation. However, these organizations are more often than not innovative—in external product terms or internal managerial ones. According to the typology used by Berne *et al*. (2019), micro-firms record three types of performance as innovative organizations: systemic, occasional, or no-innovation. The model considers various relevant dimensions, such as value proposition and customer relationships. The first type concerns structured innovation management practices and the second examines whether the organization produced any innovation during the previous three years. Berne *et al.* (2019) examined a representative number of micro-firms in a large metropolitan city and found that they overwhelmingly belong to the third type of no-innovation.

Considering that innovation leads to irreversible organizational changes, there has also been a growing body of research recently in micro-firm change management. These studies have primarily been rooted in SMEs, leading to new relevant findings. They suggest that the firm's core competence is also subject to changes (Taipale-Erävala *et al*., 2014) and that managers must readjust the organizational strategy by attempting to change their external environment (Wiesner *et al*., 2018). Some conclusions concern the more effective informal feedback methods in the change process compared to formal ones (Psychogios *et al*., 2018), the significance of intra-organizational communication as a decisive success factor in change

management (Arnaout and Esposito, 2018), as well the need for involvement of all employees in the initial stage of forced organizational change (Busse and Doganer 2018). Also, senior management plays the most critical role in the overall transformation of the firm's vision (Della Torre and Solari, 2013), while stability contributes positively to change management (Humberto Mejia-Morelos *et al*., 2013).

Nevertheless, these studies do not focus on how micro-firms manage change, as the distinction in their size is not clear. The research by Taipale-Erävala *et al*. (2021) and Rastrollo-Horrillo (2021) is much closer in this direction. The first argues that micro-firm change management must be contingent on core organizational competencies. The second concludes that micro-firms are also capable of using sophisticated management instruments. Vlados and Chatzinikolaou (2019) investigate this research gap, contending that micro-firms of less developed ecosystems do not innovate efficiently due to their physiology. This biologically oriented approach perceives organizational development as evolutionary, meaning that all dimensions of the internal environment co-evolve with each other and the broader macro-dimensions (Hodgson, 1993; Simon, 1982; Vlados, 2004; Witt, 2006). Vlados and Chatzinikolaou (*ibid.*) discern that most small firms face significant difficulties in producing, reproducing, and synthesizing their strategic, technological, and managerial trajectory (Stra.Tech.Man approach) as they cover these functions with less systematic methods that revolve around the owner's persona. This weakness makes it difficult for these firms to innovate, manage this change, and exit the crisis.[1] Table 2 captures the inherent relationship

[1] This approach of Vlados and Chatzinikolaou (2019) is adjacent to the argument of this article as it studies the evolution of small firms during the Greek crisis (circa early 2009 to late 2018). It concludes that they face competitiveness weaknesses amid the current global restructuring. Section 2.2. presents additional macroeconomic data as this article focuses on the micro-firms of a less developed Greek region.

between these dimensions as an adaptation synthesis for all socioeconomic organizations—toward a new worldwide development cycle.

Indicative literature	Theoretical contribution	Synthesizing diagnostic questions
A. Crisis (Aglietta, 2010; Mitroff, 2005; Schumpeter, 1942; Schwab, 2016; Vlados *et al*., 2018, 2022)	The crisis is an evolutionary and organic phase that causes changes in the affected system's homeostasis. This structural crisis can destroy an organization or lay the ground for amelioration. From a Schumpeterian "creative destruction" perspective, the fourth industrial revolution is accelerating nowadays due to the aging of older industries and the emergence of new ones. Today's crisis of globalization is structural, not conjunctural (or coincidental), recalibrating all worldwide socioeconomic structures. The old globalization has structurally matured and destabilized, progressively giving its place to the new globalization. This regime emerges amid continuous restructuring across all socioeconomic dimensions.	A1. What does crisis mean today? A2. What is the impact of today's crisis on the firm? A3. What are the primary causes of today's crisis? A4. What are the solutions for exiting the crisis?
B. Innovation (Berkun, 2007; Crossan and Apaydin, 2010; Grant, 1991; Porter, 1998; Schmookler, 1952; Schumpeter, 1934)	A crisis accelerates the need for a radically revamped understanding and a new way of adaptation. This process functions as an innovation springboard, which signifies new and more effective solutions than in the past. Therefore, innovation is the way to transcend the crisis. According to Schumpeter, innovation is the introduction of i) a new good or ii) a new method of production, iii) the opening of a new market, iv) the exploitation of a new source of supply, or v) the reorganization of an entire industry. This innovation is rooted in the entrepreneur's ambition and will to build a "private kingdom." Moreover, new ideas are never innovations without practical implementation, nor are minor improvements enough to create a competitive advantage.	B1. What does innovation mean? B2. What is innovation's contribution to the firm? B3. How has the firm innovated recently? B4. What are the barriers to implementing innovation?
C. Change management (Covey, 1989; Duck, 1993; Goss *et al*., 1993; Helfat, 2018; Kotler and Caslione, 2009; Kotter, 1996; Nelson, 1991; Senge, 1990; Vlados, 2019)	Innovation is the outcome of explicit or implicit managing of change. This process relates to dimensions of perception, combination, and control of organizational resources and abilities to adapt amid transforming environmental conditions. Change management refers to organizational methods for adapting to external transformations caused by internal actions from the affected system or any other socioeconomic force.	C1. What is change? C2. What is organizational adaptation? C3. What does change management mean? C4. How does the firm manage change?

Table 2. Analyzing the "crisis–innovation–change management" triangle.

According to this conceptual framework, all socioeconomic organizations can use the "crisis–innovation–change management" synthesis as a business adaptation process. Past research has examined the Stra.Tech.Man organizational physiology of small firms, posing fundamental diagnostic questions of strategy, technology, and management (Vlados and Chatzinikolaou, 2019). According to this study (*ibid*.), small firms in less developed regions have difficulty understanding and responding to changes involving strategic, technological, and managerial dynamics. This inefficiency hinders them from adapting to the new global environment (Figure 1).

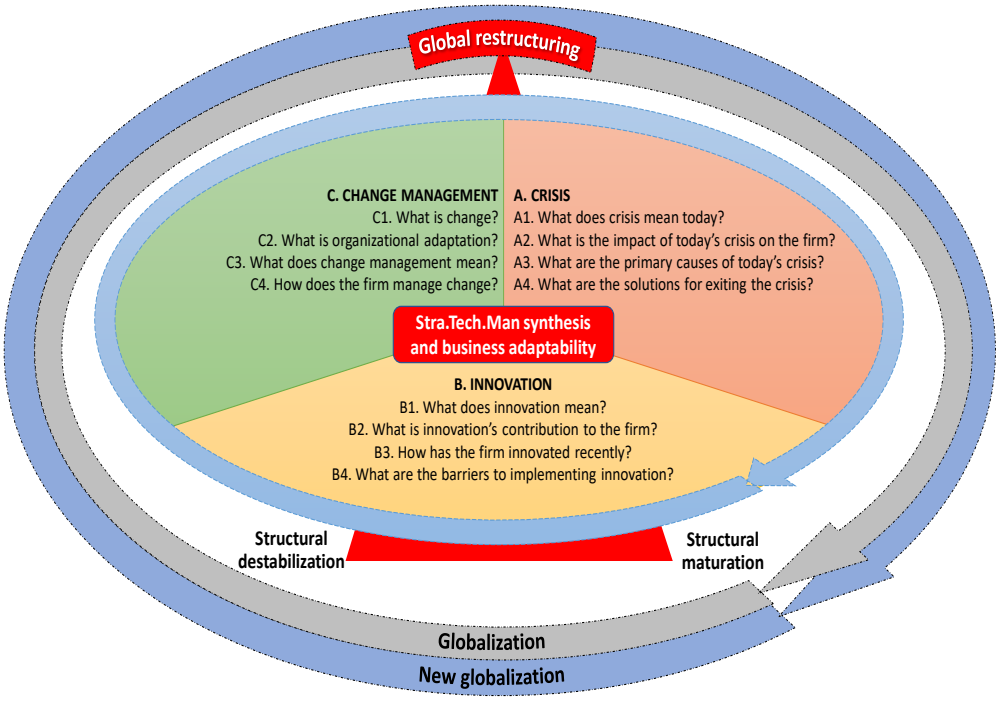


Figure 1. The conceptual framework. Based on Vlados et al. (2019).

Therefore, the crisis–innovation–change management synthesis can be the basis for a more comprehensive and straightforward diagnostic mechanism compared to the business adaptability approach based on dynamic capabilities or absorptive capacity in terms of strategy, technology, and management (Fernandes Crespo *et al*., 2021; Maiti *et al*., 2020, Teece *et al*., 1997). This paper attempts to fill this gap by exploring how less competitive micro-firms try to perceive—with varying degrees of success—the crisis–innovation–change management triangle. It examines whether their understanding of these general entrepreneurship principles is similar to the relevant scientific literature.

The remainder is structured as follows. Section 2 concerns the methodological orientation and reason for selecting the tertiary sector micro-firms of a less developed European region as a sample. Section 3 presents the main results extracted from the qualitative analysis, and Section 4 interprets the findings and discusses policy implications.

2. Methodology

This empirical research used grounded theory based on the inductive approach (Bailey, 2014; Corbin and Strauss, 2015; Lindlof and Taylor, 2002; Saunders *et al*., 2019). We selectively coded the data to extract broader axiomatic codes, concluding with a new theoretical understanding. In this approach, what already existed was less significant than emergence, a trait of all *in vivo* studies (Corbin and Strauss, *ibid.*).

Prior research in small business management has largely followed variations of this approach, such as the following: Amoako *et al*. (2021), Gomez *et al*. (2020), Grimstad *et al*. (2019), Gudeta and van Engen (2018), Khan (2018), Murthy and Paul (2017), Preedy *et al*. (2020), Senthanar *et al*. (2021). Also, we used thick description to interpret the results, as this approach means the in-depth description of human social action through the addition of subjective explanations by the researchers who collect the data and lead the research subjects to specific behaviors (Ponterotto, 2006). Given this methodological choice, the research penetrated deeper than simply recording the facts. In particular, it followed Denzin's (1989) seminal contribution by attempting to add historical elements to interpretation in addition to experience. This qualitative analysis considered the ethnographic and other characteristics of the sample.

2.1. *Qualitative analysis*

This approach allows answering questions related to "how" and "why" a situation occurs (Cater *et al*., 2017). Specifically, as stated in the Introduction, this study aims to understand how less competitive micro-firms perceive a series of development dimensions and why such firms are comparatively weaker (Figure 2).

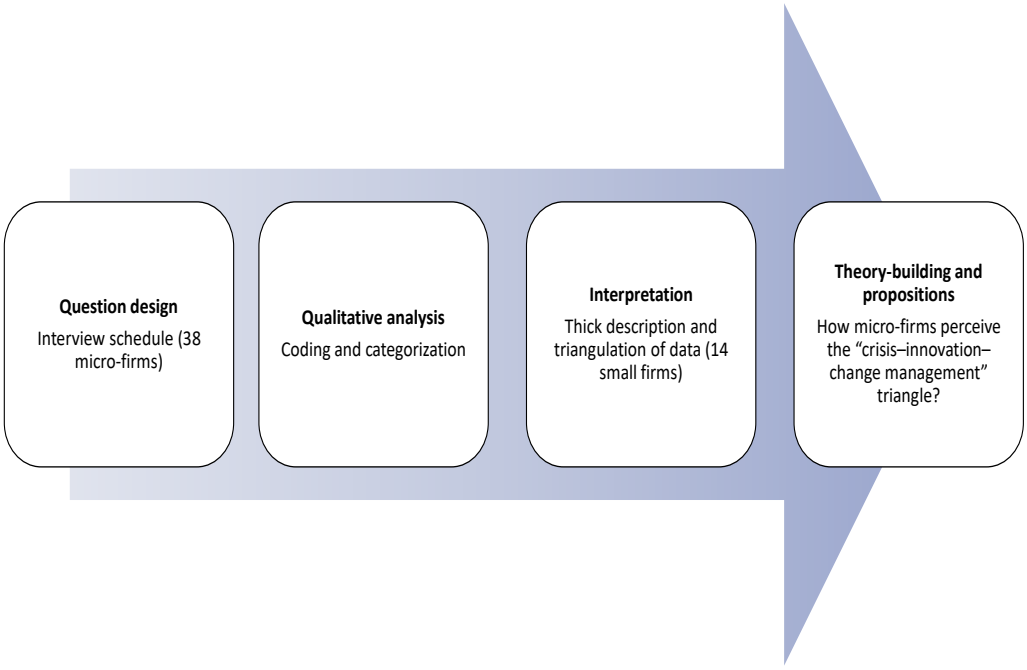


Figure 2. Research method.

This methodological approach is a variation of the qualitative analysis suggested by Lindlof and Taylor (2002). The first step was to design open-ended questions for eliciting responses from micro-firms (less than 10 employees). Building on the contributions of Sharma and Rautela (2021) and Ayodele *et al*. (2021), the interview schedule included 12 open-ended and unscaled questions about how specific business owners perceive the triangle crisis–innovation–change management. Following the related method by Yitshaki and Kropp (2016), local university undergraduates conducted the interviews in late 2017. The interviewers had to meet the owner or manager at some place within the firm, ask these questions, and sound record all answers.

The first responses' analysis showed that the respondents provided honest answers and had a counseling attitude toward the students. This finding is consistent with how Lindlof and Taylor (2002, p. 195) define the interview schedule, which "is one way to increase the reliability and credibility of data." The interviews lasted between 5 and 10 minutes on average, gradually leading to the required theoretical saturation from a sample of 38 firms. The final interviews

and thick description confirmed that the perceptions of these respondents were distant from the fundamental principles suggested in the relevant literature.

In the last step, the verbatim transcription of interviews led to coding and categorizing the 100-pages responses, as suggested by Saldana (2012). The use of spreadsheets allowed for classifying the answers into broader groups, leading inductively and subjectively to the final codes and thematic categories ("from specific to general"). The same research in small firms (11 to 50 employees) in the same sector highlighted data triangulation and confirmed the theoretical saturation from a sample of 11 respondents. Upson and Green (2020) have made a similar comparison as triangulation aims at semantic convergence by comparing two or more data sets to enhance the reliability of findings.

2.2. *The case of micro-firms in the tertiary sector of Eastern Macedonia and Thrace*

The Greek economic crisis (circa 2009 to 2018) sparked one of the longest recessions ever faced by a developed capitalist economy (Andreou *et al*., *2017*; Chryssogelos and Oxenford, 2018). A milestone of this ten-year crisis was the three successive bailout programs (from May 2010 to August 2018) under the International Monetary Fund—and the respective mechanisms created in the Eurozone (Yazgan, 2022). Other significant components were the unprecedented increase in the government debt-to-GDP ratio and unemployment (Table 3).

	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Gross domestic product (GDP), annual growth rate															
Euro area	3.0	0.4	-4.5	2.1	1.7	-0.9	-0.2	1.4	2.0	1.9	2.6	1.8	1.6	-6.3	5.4
Greece	3.3	-0.3	-4.3	-5.5	-10.1	-7.1	-2.5	0.5	-0.2	-0.5	1.1	1.7	1.8*	9.0*	8.3*
Unemployment															
Euro area	7.6	7.6	9.7	10.3	10.3	11.5	12.2	11.8	11.0	10.2	9.2	8.3	7.7	8.0	7.8
Greece	8.5	7.9	9.8	12.9	18.1	24.7	27.7	26.7	25.1	23.7	21.7	19.5	17.5	16.5	14.9
Government debt to GDP															
Euro area	66	69.7	80.2	85.7	87.6	91	93	93.1	91.2	90.4	87.9	85.8	83.8	97.2	95.6
Greece	103.1	109.4	126.7	147.5	175.2	162	178.2	180.3	176.7	180.5	179.5	186.4	180.7	206.3	193.3
Real GDP per capita															
Euro area	29,370	29,340	27,930	28,460	28,870	28,560	28,440	28,790	29,290	29,740	30,450	30,930	31,330*	29,290*	30,860*
Greece	22,500	22,370	21,350	20,150	18,130	16,940	16,630	16,830	16,900	16,890	17,110	17,430	17,760	16,180	17,590
REMTh**	14,741	15,568	15,272	14,996	13,094	12,227	11,463	11,218	11,207	11,277	11,308	11,472	11,639	-	-

Table 3. The Greek crisis in macroeconomic figures. Source: stats.oecd.org, tradingeconomics.com, statistics.gr (accessed 7.19.22). *Provisional, **REMTh = Region of Eastern Macedonia and Thrace, ***No data available

Furthermore, the Greek production system is characterized by significant structural problems such as the degrees of corruption and shadow economy (Medina and Schneider, 2018; Transparency International, 2022). Concerning present-day developments, the Greek economy seems to be recovering relatively strongly after the global recession caused by COVID-19 and the energy crisis that followed Russia's invasion of Ukraine. Specifically, it records comparatively higher growth rates than most of its Eurozone counterparts (Table 3).

Beyond these macroeconomic dimensions, the tertiary sector in Greece shows symptoms of comparative underdevelopment, focusing primarily on providing final services (business-to-consumer). These servicing forms have limited potential as the domestic income seems incapable of boosting consumer spending, at least in light of recent data (Dianeosis, 2022). In more developed economies, the tertiary sector focuses increasingly on offering intermediate services (business-to-business), bringing significant advantages to other sectors of economic activities (Mora Cortez and Johnston, 2018). This evidence reaffirms that the Greek crisis is structural and due to the relatively weak competitiveness of most Greek firms (Vlados *et al*., 2021).

The sample firms are based in the less competitive Greek region of Eastern Macedonia and Thrace (NUTS 2) and are mainly retailers and service providers. Recent studies show the comparative lag of this region, which faces structural problems of innovation and competitiveness (Annoni and Dijkstra, 2019). This region is on the southeastern borders of Europe, and, compared to other European and Greek areas, socioeconomic integration and entrepreneurial development are less potent (Boden *et al*., 2015; Vlados and Chatzinikolaou, 2020). The GDP per capita is well below the European average (Table 3), the demography includes more minority populations than most Greek regions, and, in socioeconomic terms, it shows high comparative costs for its citizens and firms (Arieli, 2019; Boden, 2017).

Concerning morphological characteristics of this business ecosystem, the regional economy faces structural issues as most firms belong to the service sector with a narrow local focus and customer base—the Greek GDP comes from services by 80% approximately (CIA, 2022). The

area's relatively high export performance seems to be linked to the secondary sector—the region is ranked 7th out of 13 in export volume (SEVE, 2018). Also, the number and turnover of tertiary sector firms in this region fell by approximately 20% between 2011 and 2018 (own processing of the following data: Hellenic Statistical Authority, 2018).

Amid this Greek crisis, our team visited several local firms to gain insights into how they adapt. The non-probability sampling was the method to build the representative sample to achieve theoretical saturation, meaning the point where no new thematic categories emerge from the data (Guest *et al*., 2006). Also, we had prior knowledge that helped us classify these micro-firms as less dynamic and adaptable due to living and working in the area for several years. Table 4 lists and assigns a number to the sample of 38 micro-firms (MF) and 14 small firms (SF).[2]

Micro-firms (1-10 employees)		**Small firms (11-50 employees)**	
MF1	Advertisement and Internet services	SF1	Hotel
MF2	Food services	SF2	Café-bar
MF3	Night club	SF3	Café-bar
MF4	Drug store	SF4	Food and café services
MF5	Café services	SF5	Advertisement services
MF6	Drug store	SF6	Steakhouse (grill)
MF7	Food and café services	SF7	Household electrical appliances and furniture retail and services
MF8	Decoration services		
MF9	Grocery		
MF10	Retail trade of decoration items	SF8	Everyday apartment building expenses and brokerage services
MF11	Convenience store		
MF12	Hair salon		
MF13	Driving school	SF9	Café-bar
MF14	Hair salon	SF10	Restaurant
MF15	Fitness center	SF11	Café-bar
MF16	Accounting services	SF12	Food and café services
MF17	Trading of construction materials	SF13	Hotel
MF18	Retail trade of decoration items	SF14	Hotel
MF19	Internet café		
MF20	School of foreign languages		
MF21	Bakery		
MF22	Trading of fertilizers, tools, animal feed, and pesticide		
MF23	Trading of timber, construction materials, and industrialized materials		
MF24	Retail trade of housekeeping items		
MF25	Trading of household electrical appliances		
MF26	Trading of beauty, aesthetics, and personal care items		
MF27	Fitness center		
MF28	Accounting services		

[2] We promised all respondents not to publish any personal information. Also, all raw data is in spreadsheets and available to interested researchers upon request.

MF29	Night club and café-bar
MF30	Drug store
MF31	Retail trade of perfume
MF32	Wholesale of electrical equipment
MF33	Real estate agency services
MF34	Candy shop
MF35	Steakhouse (grill)
MF36	Barbershop
MF37	Retail trade of smoking-tobacco items
MF38	Postal services and courier

Table 4. The sample firms per size and business (based on the owner's description).

All the sample micro-firms are less dynamic; they have a purely local focus even though they are based in central areas of the region's large cities. Also, most are family-run and have been operating for more than ten years. About half employ five or fewer employees (in some cases, the firm occupies only its owner). As for the small ones, nearly half have more than 30 employees and are more dynamic. These larger firms are more extroverted, with higher turnover, and some attract an international clientele. Therefore, primarily the responses of these firms led to differentiated categories.

3. Qualitative analysis results

Table 5 presents the generated coding separated into micro-firms and small firms. In the various codes, the first two letters concern the analyzed dimension ("CR" for crisis, "IN" for innovation, and "CM" for change management)—"MF" and "SF" symbolize the firm's type (micro-firm and small firm). The last letter shows the descending order in the code's occurrence. Also, Table 5 is the point of reference for the subsequent analysis, which cites the different codings frequently.

Micro-firms	Small firms
Crisis perception, impact upon the firm, and solutions	
{CRMF-A} Reduced demand and turnover	{CRSF-A} The crisis is due to both poor public management and business inefficiency
{CRMF-B} Problematic management by the national governance	{CRSF-B} Both the customer's purchasing power and the firm's turnover have decreased
{CRMF-C} The need for investments (public principally)	{CRSF-C} Economic development by attracting private and public investment
{CRMF-D} Increase in taxes and costs	{CRSF-D} The crisis does not affect the firm, functioning as a motor for quality improvement
{CRMF-E} Everyone is facing the same problem	{CRSF-E} The problem is socioeconomic and multilevel
{CRMF-F} Insufficiency on the part of firms	{CRSF-F} I do not know
{CRMF-G} Excessive borrowing	
{CRMF-H} Cut costs	
{CRMF-I} Battle for survival and striving for the best	

{CRMF-J} Increased uncertainty and competition	
{CRMF-K} We work more hours compared to the past	
{CRMF-L} Lack of knowledge	
{CRMF-M} Layoffs	
Innovation perception, specific innovations, and obstacles	
{INMF-A} Innovation is something that improves the firm through new products or improvement of existing ones	{INSF-A} Creating a new product or method and improving existing ones
{INMF-B} It is not clear, and the firm probably does not innovate	{INSF-B} The firm's liquidity
{INMF-C} Liquidity problem	{INSF-C} We face difficulties
{INMF-D} Innovation attracts new customers, increases revenue, and reduces costs	{INSF-D} There are no obstacles
{INMF-E} Institutional barriers (high taxation, national insurance contributions)	{INSF-E} Local market barriers
{INMF-F} Innovation is primarily new ideas	
{INMF-G} Market barriers	
{INMF-H} Innovation concerns specialized applications	
{INMF-I} Cultural barriers	
{INMF-J} Problems in human resources	
{INMF-K} Uncertainty	
{INMF-L} There are no obstacles, and we must not stick to anything old	
Change perception and management	
{CMMF-A} Adapting to new demand conditions and improving the firm	{CMSF-A} Business improvement
{CMMF-B} Reduced income and expenses	{CMSF-B} Continuous adaptation
{CMMF-C} Managing change is difficult	{CMSF-C} Forced adaptation
{CMMF-D} The firm's institutional framework changes	{CMSF-D} Better trained staff
{CMMF-E} The firm does not need to change	{CMSF-E} It is not clear
{CMMF-F} Increased competition	{CMSF-F} Adapting to change is time-consuming but beneficial
	{CMSF-G} The customers' purchasing power is reduced
	{CMSF-H} Staff reduction
	{CMSF-I} Increase in demand

Table 5. Coding of responses.

The higher codes in the three lists have more weight than the lower ones, leading to specific thematic categories.[3] The following subsections correspond to this emerging categorization, generalizing upon the findings.

[3] The codes in Table 5 could be the basis of future mixed research in the form of closed and open-ended questions on a graded scale—for example, from -2 to +2. In this new approach, the CRMF-A code would examine, for example, the respondents' perception of "the extent to which the crisis means reduced demand and turnover." This restructured survey could concern a large representative sample of micro-firms in less dynamic business ecosystems.

3.1. *Micro-firms*

3.1.1. The crisis is primarily external

Most less competitive micro-firms reported that reduced demand or revenues and a forced cut in their spending signal the advent of a crisis (A, H, M). Although they also mentioned other dimensions that drastically transform their internal environment, they presented the need for change at the national government level as imperative and demanded an increase in public investments (B, C, E, I, K). Other external dimensions, such as the current tax increase, the excessive borrowing of citizens, and the intensification of competition, showed a conjunctural perception of the crisis (D, G, J). There is a minority that, exceptionally and under certain conditions, would be willing to understand the more profound crisis causes (F). Also, some firms showed a complete lack of knowledge and inability to analyze. Finally, the owner of a firm that trades household electrical appliances and employs nine people on a central spot in one of the largest cities in the region responded that:

MF25: The crisis has affected the whole world and our business. The government must increase its spending and reduce taxes.

3.1.2. Innovation is good but not for us

Progress through organizational novelty sounds auspicious to these less adaptable micro-firms (A, D, L). However, when asked to report specific innovations (B), they appeared ambiguous and uncertain. These less flexible micro-firms tended to focus on various obstacles— primarily external—that prevent them from implementing novel solutions (E, G, I, K). They argued that their liquidity problem was mainly related to the national government's mismanagement and the inability to provide subsidies and low-interest loans (C). A minority reported issues related to human resources (J) and that a good idea could be innovation (F). Also, some firms stated that innovation could only concern specialized applications without further elaboration (H). As

an example, a beauty, aesthetics, and personal care retailer on a major city street with two employees responded that:

> *MF26: Innovation and creativity are extremely important. We have not been able to innovate because we belong to the specific category of small and medium-sized enterprises. It is not that we do not want it. The idea is there, but the dynamics are limited, resulting in the inability to innovate. The reasons are obvious: economic.*

3.1.3. Remarkable endurance despite adverse conditions

These less competitive micro-firms reported that they adapt to change despite the challenging conditions of their external environment, striving to improve the business (A, D, F). This adaptation and improvement appeared to follow primarily a course of downsizing the firm and diminishing ambitions, showing a willingness to keep the business alive and endure despite the lack of profits (B). No firm could describe any systematic ways it manages change (C). Also, some firms argued that nothing is changing, having earlier stated that they adapt to emerging conditions (E). A five-employee school of foreign languages located in a large city center shows the overarching pattern:

> *MF20: Every owner must learn to behave like an employee. We have reduced the number of employees. We did this because I started working much harder, I would say exhaustively.*

3.2. *Small firms*

3.2.1. The crisis is due to internal and external causes

Small firms appeared to have a deeper understanding of the crisis. Unlike micro-firms that primarily highlighted poor public administration, small firms declared business inefficiency as another significant determinant (A, B, D, E). They did not mention only public investments as a growth springboard but mainly a mixture of entrepreneurial and institutional development

(C). A minority claimed they did not know enough about the crisis (F). A 13-employee firm located in a central city location and manages everyday apartment building expenses and offers brokerage services stated that:

SF8: The tax system must be improved. Also, vocational training schools need upgrading because there is a shortage in some professions and blue-collar workers.

3.2.2. Innovation is a complex concept

In most dynamic small firms, innovation is an intricate but attainable process. These organizations also reported changes in organizational methods besides introducing new products or improving existing ones (A). The emergence of various barriers showed that the entrepreneurial climate in the region is still generally in a downward spiral (B, C, E). Also, some firms reported that they do not face any problems in their innovative development (D). A fairly extroverted hotel that employs 32 to 50 personnel depending on the season responded that:

SF13: Implementing innovative ideas will help the firm attract new customers and better serve the existing ones. We can achieve this by purchasing and utilizing new machines and creating modern service methods.

3.2.3. Change management is primarily an improvement mechanism

The most dynamic small firms stated that managing change means improving the business and a continuous adaptation effort, which also is mandatory (A, B, C, F). The pattern that emerged from the responses was that, although not fully understood, change management leads to various organizational improvements (D, E, G, H, I). A restaurant owner located in a downtown area of the region and employs 19 people said that:

SF12: Two types of factors contribute to change. The external can be economic, political, and social. The internal include customer needs and the firm's philosophy.

4. Discussion

Our findings suggest that less dynamic and adaptable micro-firms perceive certain fundamental principles of entrepreneurship differently from systematic approaches to business organization theory (Berrone *et al*., 2014; Faherty & Stephens, 2016; Lin *et al*., 2020; Rigg *et al*., 2021). In particular, regarding the crisis–innovation–change management triangle, the following points summarize this distance in perception.

A. These individuals in less competitive MFs seek salvation primarily through government intervention. They see the crisis as a quasi-exogenous phenomenon that the national government can mitigate if it increases spending, lowers taxation, and takes sectoral protection measures. This understanding is relatively far from the existing literature that examines the structural nature of crises and their irreversible arrival due to the fulfillment of evolutionary conditions. As relevant literature suggests, a structural crisis manifests as a series of pressing problems that eventually lead to a drastic overthrow of the previous status (Aglietta, 2010; Mitroff, 2005; Schumpeter, 1942; Schwab, 2016; Vlados et al., 2022).

B. Although these weaker firms—in financial, competitive, and organizational terms—recognize the importance of innovation, they define it as extremely difficult and quasi-unattainable. According to relevant literature, innovation is the only way to transcend the crisis and enter a new development spiral. Thus, the lack of innovation confirms why these firms are less profitable and not as competitive (Berkun, 2007; Crossan and Apaydin, 2010; Grant, 1991; Porter, 1998; Schmookler, 1952; Schumpeter, 1934).

C. The answers from these individuals show that they cannot answer sufficiently to change management concepts. These micro-firms do not seem to implement systematic change management mechanisms and procedures, even though the owners appear willing to improve

their business. As the next section discusses, this paradoxical improvement happens by limiting their profits and ambitions. However, relevant literature suggests that change management is an overarching process relating to dimensions of perceiving, combining, and controlling organizational strengths for adapting to transforming conditions (Covey, 1989; Duck, 1993; Helfat, 2018; Goss et al., 1993; Kotler and Caslione, 2009; Kotter, 1996; Nelson; 1991; Senge, 1990; Vlados, 2019).

4.1. *A synthesizing thick description*

The respondents appeared to embellish their answers somewhat, fearing possible negative criticism. Most respondents gave one-word responses or none whenever the interviewers attempted a laddering approach to discern the "why." One explanation for this behavior is the absence of knowledge or deeper processing related to the triangle crisis–innovation–change management.

The small-firm owners appeared more authoritative and receptive to answering laddered questions. One explanation for this more comprehensive processing capacity is the firm's breadth. This interaction proved that business size makes these individuals more extroverted and obliged to consider more factors. The sample owners of developed small firms have a greater capacity for complex thinking than owners of less competitive and flexible micro-firms. Another explanation for this lies in the different levels of education, as the less dynamic sample micro-firms do not employ as many people with a postgraduate—primarily—or undergraduate degree as small ones. The owners of sample less competitive micro-firms are predominantly middle-aged people who appear detached mainly from fundamental scientific developments in entrepreneurship.

These micro-entrepreneurs do not emphasize innovation as a determinant of organizational survival—the responses regarding how the firm has innovated confirm this finding. Also, phobic introversion accompanies this lack of knowledge, manifested several times as a "schooling attitude" toward the younger interviewers. Although a standpoint open to learning

new relevant techniques and methods from the trainees could be beneficial, many entrepreneurs showed either a form of arrogance or a complete resignation. In other words, their behavior oscillated between two extreme views as an attempt to cover up weaknesses due to fear—micro-firm owners in relatively closed societies in Greece often appear introverted (for the profile of Greek export firms, see Dianeosis, 2018). These entrepreneurs' knowledge is usually limited to the direct competitor "across the street." Their ambitions significantly differ from Schumpeter's (1934) "private kingdom," as the majority perceive improvement as an act of asceticism by limiting business profits and personal needs. Amid this paradoxical introverted behavior, these owners seemingly blame external factors, stubbornly refusing to understand the weaknesses of their internal organizational environment.

Therefore, these findings seem to confirm that most sample micro-firms are "monad-centered" (revolve around the owner's persona, the "monad") in terms of organizational physiology Stra.Tech.Man. According to the relevant typology of Vlados *et al*. (2021), the three forms of entrepreneurship in Greece are "monad-centeredness," "massiveness," and "flexibility." The vast majority of firms seem adjacent to "monad-centered" physiology. This fact appears to create competitiveness problems in the Greek economy as these organizations follow instinctive forms of strategy, sporadic technological choices, and management methodologies exhausted in the daily market experience. Based on this approach, the findings seem to validate why these identified atypical forms of organization and management are indicators of different evolutionary—physiological—trajectories in firms (Helfat, 2018; Hodgson, 1993; Vlados, 2004; Witt, 2006). Also, they presumably verify that systematic–massive management models (see Taylorism) have limited scope in less dynamic socioeconomic organizations (Nelson, 1991; Simon, 1982).

The selection of this sample appears to some extent ideal and representative as the responses were uniform, leading relatively quickly to theoretical saturation. We had prior knowledge of the region that helped us classify these firms as less dynamic and adaptable—approximately two to three years since the last data collection, several of these have gone out of business or

are inactive. This study appears to reach a similar finding to Berne *et al*. (2019), as most micro-firms in the sample belong to the no-innovation type. In essence, the crisis–innovation–change management triangle is a blind spot for these less competitive micro-firms as they ignore its significance as an adaptation mechanism—contrary to several direct competitors (Figure 3).

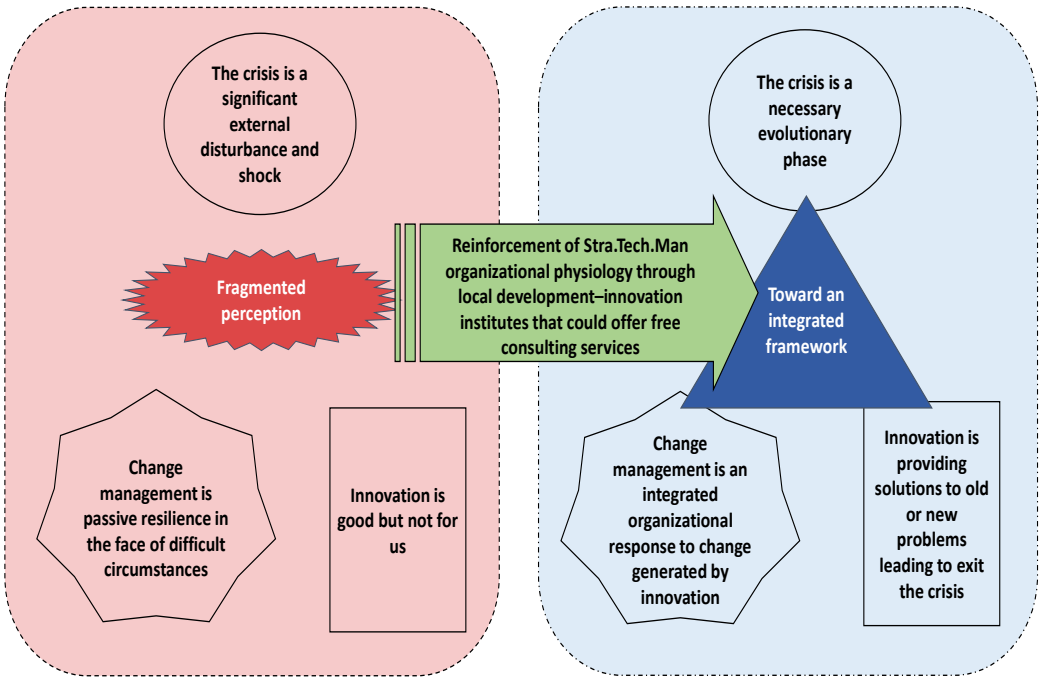


Figure 3. A gap between the fragmentary perception of micro-firm owners and the relevant scientific theory.

The gap between these individuals and the respective theoretical development in entrepreneurship is evident. How can policymakers help remedy this fragmented perception that breeds weaknesses? How can they get these entrepreneurs to see the crisis as an inevitable evolutionary phase that organizations overcome by implementing innovation and change management mechanisms? As the next section presents, the solution to this problem can come through reinforcing the Stra.Tech.Man physiology (more effective synthesis of strategy–technology–management) with the care of related public policies.

4.2. *Social implications*

No matter how selfish you think man is, it's obvious that there are some principles in his nature that give him an interest in the welfare of others, and make their happiness necessary to him, even if he gets nothing from it but the pleasure of seeing it. [Adam Smith, 1759, The Theory of Moral Sentiments]

When they have limited ambitions and inadequate business organization knowledge about the meaning and process of innovation, the less competitive micro-firms of less developed ecosystems highlight the external environment's unfavorable conditions to offset their weaknesses. Reductio ad absurdum, this finding could not be untrue. If these firms had greater ambitions and relevant knowledge, they would be primarily interested in improving, aiming for sustainable and sufficient profitability, as some of the slightly larger firms in the same sector and area seem to be doing.

What can policymakers do for these micro-firm owners to face the intensified new global competition? Although the added value for these firms is negligible in financial and macroeconomic terms, they are a significant social group. Beyond reasons concerning economic development, the classical liberal perspective suggests social empathy as the glue of societies (Smith, 1759, 1776). Thus, the state should build hospitals if a community has many weak and sick people. These firms resemble ill persons with chronic conditions that only official hospital structures can treat. Based on the reluctance of these research subjects to cultivate their entrepreneurship knowledge, it seems unrealistic that these owners themselves would seek to pay the necessary high price for business consulting in the future. An ideal solution would be to build public entrepreneurship clinics to provide these weak organizations—in competitiveness, flexibility, and innovativeness—with free preliminary or in-depth counseling.

This study introduced an initial diagnosis framework for less dynamic micro-firms based on the crisis–innovation–change management triangle. However, policies that fail to diagnose the

internal organizational environment are ineffective in reinforcing SMEs as they focus exclusively on macro–meso interventions that favor larger enterprises (Chatzinikolaou *et al*., 2021). Entrepreneurship reinforcement policies must perceive the environment as a macro–meso–micro synthesis (Chatzinikolaou *et al*., *ibid*.). This local or sectoral intervention must diffuse modern entrepreneurship methods to firms, combined with the provision of financial facilities (Figure 4).

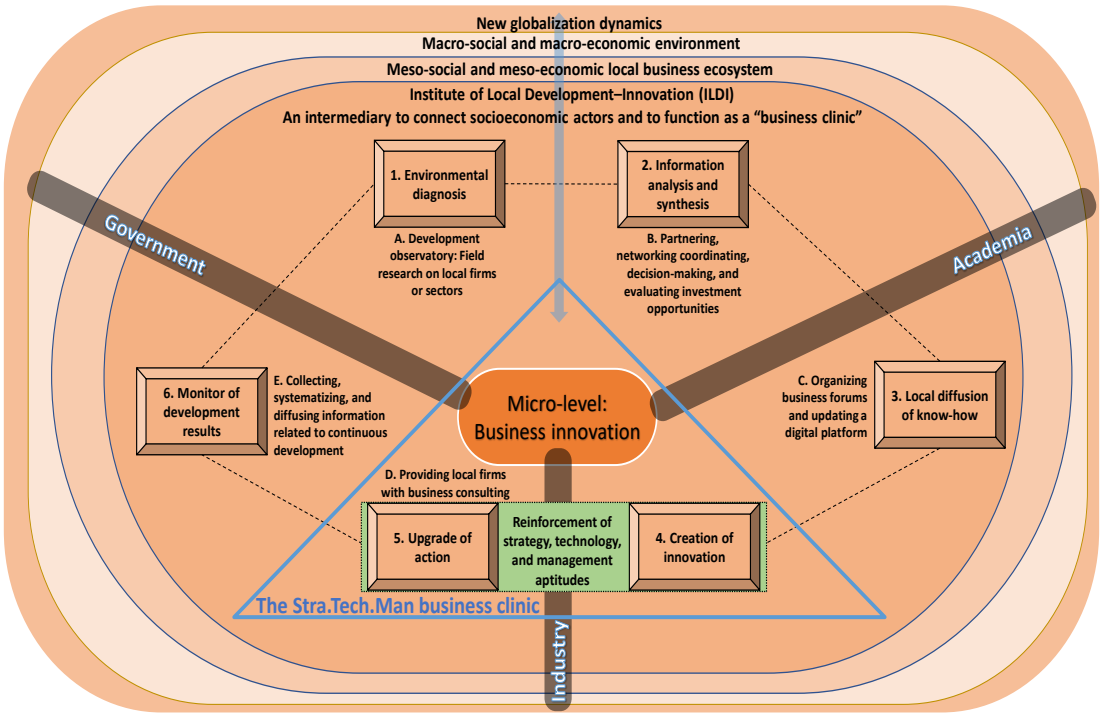


Figure 4. The Institute of Local Development–Innovation (ILDI): a framework for reinforcing entrepreneurship in an emerging regional economy.

This policy proposal is not yet another vague networking of meso-actors. The Institute of Local Development–Innovation (ILDI) could aim to strengthen the Stra.Tech.Man organizational core (Vlados and Chatzinikolaou, 2019). The ILDI could provide free consulting services as an intermediary to bring together socioeconomic actors and function as a "business clinic" that diagnoses and reinforces innovation in local firms. These different institutions are usually uncoordinated in emerging economies. First, triple-helix structures (academia, industry, and government) cannot generate innovation if they do not stimulate the entire macro–meso–micro socioeconomic system as business progress changes the structures from within. Second, there

is a tendency toward creating meso-institutions such as incubators, technology or science parks, accelerators, spin-offs, research centers, co-working spaces, technology transfer offices, chambers of commerce, or observatories. These bodies usually occupy specialized niches in the economy as reinforcing weaker organizational structures seems detrimental to public or private financial resources. Also, banking institutions, business angels, and venture capitalists follow similar patterns and focus on specific high-value socioeconomic segments (Rodríguez and Navarro-Chávez, 2015; Sun and Cao, 2018).

The ILDI policy proposal could emphasize problematic micro-firms like the ones examined here. This institution could resemble similar interventions such as the Training and Enterprise Councils (TECs), a policy developed in the UK aimed at evolving the national workforce organization into a small-business support body (Adam-Smith and McGeever, 1995). The TECs were private entities established locally, contracting with the government to provide services related to strengthening local employability. Also, the TECs financed small firms to contract with external business consultants (Boocock *et al*., 1994).

The ILDIs' vision attempts to restore the now abandoned work of TECs (Peck and Jones, 1995). Different emerging regional–national economies could promote this policy through private-public partnerships. Also, the ILDIs could set five enduring internal organizational goals. First, they could be a development observatory for the regional–local economy. Second, they could make decisions to coordinate initiatives between innovation-related actors, aiming to find investment opportunities. Third, they could emphasize local knowledge dissemination through periodic forums and relevant digital platform updates. Fourth, they could provide—like the TECs—free advising services to local firms by funding external consultants. The goal could be to reinforce these firms' perception of the crisis–innovation–change management triangle, based on improving the Stra.Tech.Man innovation core. Fifth, they could monitor their results by publishing periodic progress reports based on these five goals.

4.3. *Limitations*

Inevitably, this research is subject to barriers that require theoretical elaboration. The paper focused on less competitive micro-firms in a less developed European region; therefore, it does not directly apply to other areas with significant shortages in critical infrastructures, such as Sub-Saharan Africa. Furthermore, we only examined micro-firms in the tertiary sector. However, the emergence of similar patterns in the primary and secondary sectors is likely and should be explored. Also, further research into the different entrepreneurship physiologies of business ecosystems in Greece and abroad seems imperative. This enhanced inquiry could lay the ground for further distinguishing the various business typologies and strategic, technological, and managerial qualities besides company size (Vlados, 2004). Finally, future research could reevaluate how these less dynamic micro-firms have changed due to the COVID-19 pandemic and today's energy crisis.

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